SPEED DETECTION RADAR SYSTEM USING ARDUINO UNO AND IR SENSOR

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This project focuses on designing a low-cost speed detection radar using two IR sensors, an Arduino Uno, a 16x2 LCD display, a buzzer, and an LED. The project records the time an object passes between two sensors placed at a fixed distance, calculates the speed in both cm/s and km/h, and displays it on an LCD screen. An alert system is triggered if a set speed limit is crossed.

Components Required

1. Arduino Uno
2. IR Sensors (2)
3. 16x2 LCD Display (I2C)
4. LED
5. Buzzer
6. Breadboard
7. Jumper Wires

Connections :

IR Sensor 1 Output -> D2

IR Sensor 2 Output -> D3

LED -> D6 (with 220 ohm resistor)

Buzzer -> D7

Speed Detection Radar Using IR Sensors and Arduino Uno

LCD SDA -> A4, SCL -> A5

5V and GND connections as needed

Code :

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);



#define IR1_PIN 2

#define IR2_PIN 3

#define BUZZER_PIN 4

#define LED_PIN 5

const float distanceCM = 50.0;

const float speedLimit = 20.0;

unsigned long startTime = 0;

unsigned long endTime = 0;
```

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```
bool objectDetected = false;

void setup() {

 pinMode(IR1_PIN, INPUT);

 pinMode(IR2_PIN, INPUT);

 pinMode(BUZZER_PIN, OUTPUT);

 pinMode(LED_PIN, OUTPUT);

 digitalWrite(BUZZER_PIN, LOW);

 digitalWrite(LED_PIN, LOW);

 lcd.init();

 lcd.backlight();

 lcd.setCursor(0, 0);

 lcd.print("Speed Radar");

 delay(2000);

 lcd.clear();
```

Speed Detection Radar Using IR Sensors and Arduino Uno

```
}

void loop() {

 int sensor1 = digitalRead(IR1_PIN);

 int sensor2 = digitalRead(IR2_PIN);

 // Object breaks IR1

 if (sensor1 == LOW && !objectDetected) {

 startTime = millis();

   objectDetected = true;

 }




 if (sensor2 == LOW && objectDetected) {

 endTime = millis();

 unsigned long timeTaken = endTime - startTime;

 float timeSec = timeTaken / 1000.0;
```

Speed Detection Radar Using IR Sensors and Arduino Uno

```
if (timeSec > 0) {

float speedCmps = distanceCM / timeSec;

    float speedKmph = speedCmps * 0.036;



     lcd.clear();

     lcd.setCursor(0, 0);

     lcd.print("Speed:");

     lcd.print(speedKmph, 2);

     lcd.print("km/h");



     lcd.setCursor(0, 1);

     lcd.print("Speed:");

     lcd.print(speedCmps, 2);

     lcd.print("cm/s");
```

Speed Detection Radar Using IR Sensors and Arduino Uno

```
    digitalWrite(LED_PIN, HIGH);

    if (speedKmph > speedLimit) {

      digitalWrite(BUZZER_PIN, HIGH);

      delay(1000);

      digitalWrite(BUZZER_PIN, LOW);

    }

    delay(2000);

    digitalWrite(LED_PIN, LOW);

  }

  objectDetected = false;

 }

}
```

Speed Detection Radar Using IR Sensors and Arduino Uno

Working Principle

Two IR sensors are placed 10 cm apart. Arduino records start and end times as the object crosses sensors.

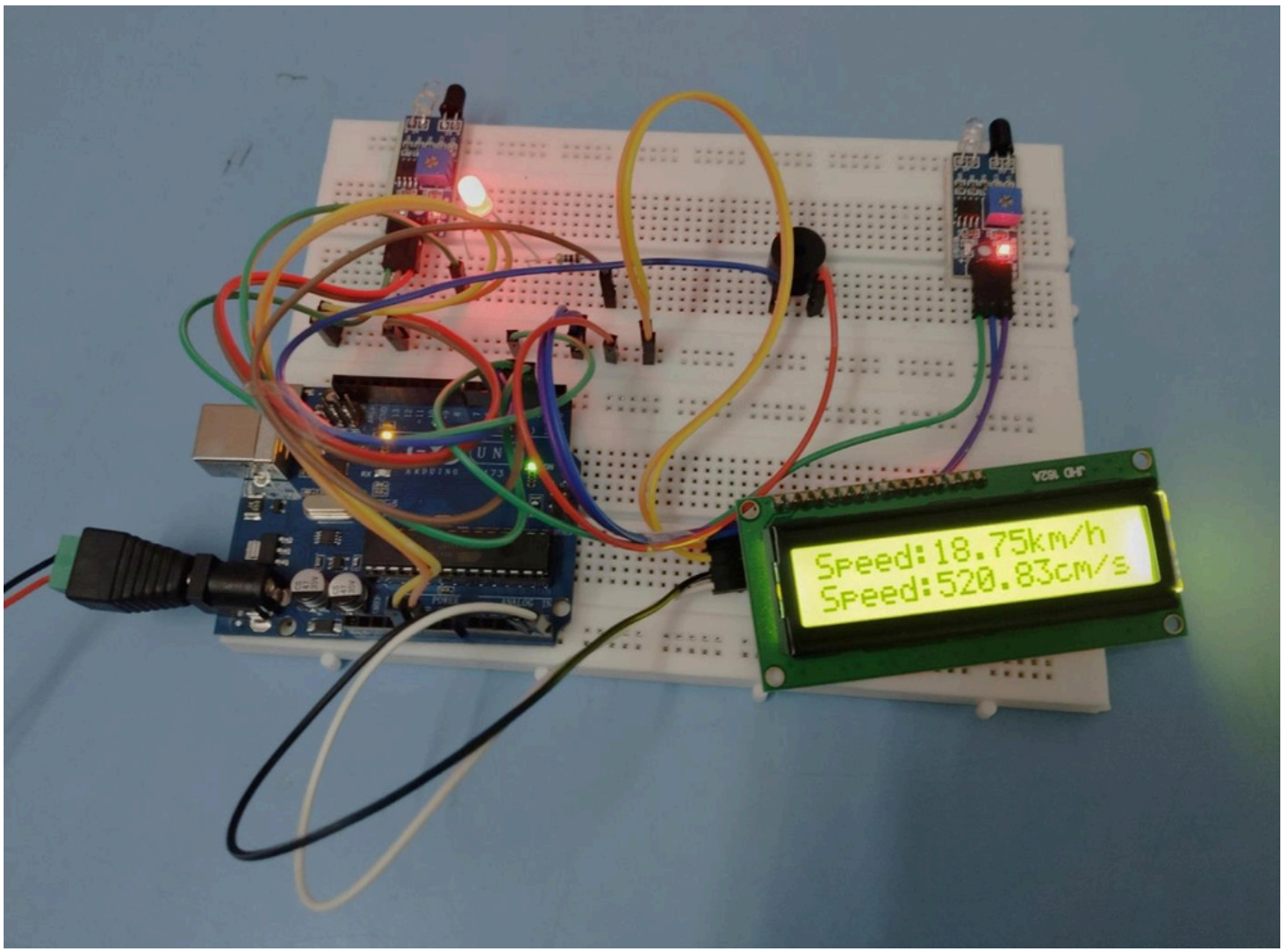
Speed is calculated:

- Speed (cm/s) = Distance / Time

- Speed (km/h) = Speed (cm/s) x 0.036

The LCD displays speed in cm/s and km/h. If speed exceeds a threshold, LED and buzzer are triggered.

Implementated Design:



Speed Detection Radar Using IR Sensors and Arduino Uno

Applications

- Checking speed of moving objects.
- Speed monitoring for toy cars and robotics.
- Educational experiments.

Advantages

- Low-cost and simple design.
- Easy to use and portable.
- Useful for small industrial automation tasks.

Conclusion

The Speed Detection Radar project successfully measures object speed in cm/s and km/h, using basic components like IR sensors and Arduino Uno. It serves as a basic foundation for more advanced systems.